Forecasting where animals move next

One animal’s history.
Many plausible futures.

Probabilistic movement forecasts from 1 hour to 90 days

108 training taxon labels

Four predictors, one ensemble

Measured local gains

Research objective: frontier migration forecasting

Evidence: immutable corrected cache; current packaged ensemble; matched local development panel.

The dataset and the inputs

34.8M
reviewed position observations

799,681
training forecast origins

108
training taxon labels

Public animal tracking studies

131 study labels in TRAIN

Terrestrial animals, birds and marine examples

Current leading inputs

Observed GPS + timestamps + species

Derived season, velocity and daylight

108 labels = 107 species names + Testudinidae (family). Observed future positions are never interpolated.

Sources: corrected manifest and source preflight; migration_repaired_features.py. Forecast origins are not independent animals.

From observed history to plausible futures

Observed history

29 lag slots + masks

Species + 22 derived features

Minutes through one year

Four joint predictors

Causal Transformer encoders

Direct location × 2

Goal route × 2

Joint future distribution

8 components per predictor

Weighted ensemble

64 sampled paths × 9 horizons

≈35M total parameters

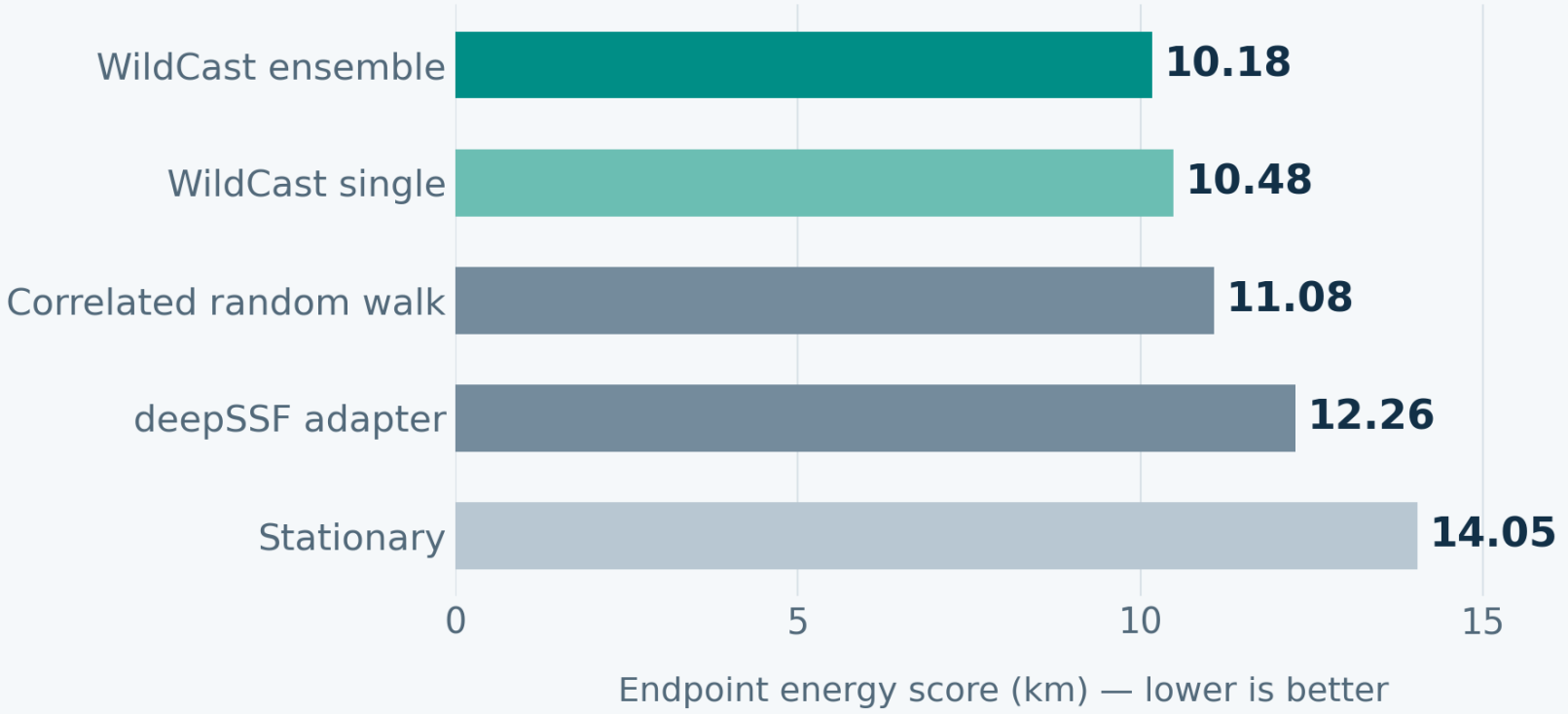
A predictor is selected once per complete sampled route

Sources: migration_model.py; migration_ensemble.py; packaged member configurations and weights in evidence.json.

MEASURED COMPARISON

Better scores on the matched local panel

4,096 identical origins • 1h / 12h / 1d • 64 samples



8.1%
lower than CRW

17.0%
lower than deepSSF

Energy evaluates the full predicted distribution. Development results are averaged across two validation splits.

Source: migration-public-panel-v1/metrics.json. Local adaptations; ensemble probabilities fitted on development; no official MoveBench rank.

Why combining predictors can help

History and season

Connect recent movement with annual context

Several possible futures

Represent different destinations and uncertainty over time

Complementary predictors

Combine direct forecasts with a destination-based bias

2.9% **lower energy than our strongest single predictor**

The gain is concentrated in unseen animals. The held-out-year difference is small and uncertain.

Sources: matched public-panel scores and paired cluster bootstrap; model source. Mechanisms below are hypotheses, not isolated causal findings.

Constraints and the next improvements

Current constraints

Environment and demographics missing

Uneven species and regional coverage

Movement labels mix migration and residence

Independent final test still pending

Priorities

1 Freeze independent public comparisons

2 Add dated environmental and trait data

3 Expand independent marine populations

4 Validate calibration and transfer

Today: promising local gains. Next: demonstrate reliable superiority on independent data.

Sources: current input code, metadata proposal, context trials and evaluation scope. Frontier superiority remains unestablished.

Inputs and outputs in detail

Model inputs

29 × 10 history features + 29 masks

22 derived static features

108 taxon tokens + unknown token

At least 8 prior fixes spanning ≥6h

Model outputs

9 horizons: 1h, 12h, 1, 3, 7, 14, 30, 60, 90d

Samples from a joint route distribution

Location and uncertainty at each horizon

No learned animal-ID embedding

Local RTX 4060 Laptop: 108 tracks × 64 futures × 9 horizons in ≈41 ms

Warm model and prepared inputs; coordinate conversion included; raw-track preparation and loading excluded.

Sources: migration_repaired_features.py; packaged configs; inference benchmark; data manifest.

Dataset split and coverage

TRAIN	**799,681**	Cutoff through 2020
Validation: unseen animals	**201,563**	Conservative identity exclusions
Validation: held-out year	**43,896**	2021
Reserved assessment	**204,266**	Targets remain sealed

Marine examples: green and hawksbill turtles, grey seals, sperm whales and seabirds.

144 study labels have cases; 131 have TRAIN cases. Coverage is uneven and is not a sample of all animal migration.

Sources: corrected cache manifest, preflight, study-inventory.csv and taxon-inventory.csv.

Architecture and evaluation details

Predictor	Width / layers	Ensemble share
Direct compact	128 / 2	**8.6%**
Direct scaled	512 / 6	**45.6%**
Goal route compact	128 / 2	**14.2%**
Goal route scaled	512 / 6	**31.6%**

TRAIN-only normalization; square-root taxon sampling; NLL + energy optimization.

Full development: 245,459 origins, 1,734,789 observed targets and 64 draws at nine horizons.

Sources: packaged checkpoint configurations; migration_train.py; migration_evaluate.py; migration_ensemble.py.

What the score means and what it proves

Energy score

Rewards samples near the observed future, with a correction for forecast dispersion. Lower is better; units are kilometres.

Mean point error

Measures the miss of the forecast mean. It answers a different question from the full distribution score.

What we can claim

Better energy on the tested local panel

What still needs evidence

Independent transfer, official benchmark ranking and global frontier superiority

Evidence: exact local report hashes in evidence.json. Method code: github.com/cifkao/moveformer; github.com/swforrest/deepSSF_package.